FACULTY OF ENGINEERING
AND ARCHITECTURE

# Development, simulation & implementation of precoding techniques for satellite links


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Master's dissertation submitted in order to obtain the degree of Master of Science in Computer Science Engineering

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by

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Ghent University

Faculty of Engineering and Architecture

Department of Telecommunications and Information Processing

**Abstract** — Laboris magna irure quis sit et veniam voluptate duis elit esse do excepteur. Officia laboris adipisicing laboris duis ut cillum aliquip. Do veniam aute non do cillum sit pariatur consectetur adipisicing qui. Lorem Lorem veniam minim exercitation cupidatat tempor do aute dolore. Nostrud adipisicing tempor quis qui fugiat sint quis ut aute eiusmod Lorem enim veniam. Elit labore Lorem do proident amet anim exercitation culpa enim cillum qui. Aliquip reprehenderit commodo labore minim laborum mollit ut veniam in irure. Minim sit in non aliqua ut elit veniam irure pariatur non velit consequat. Cupidatat veniam occaecat culpa eu irure aute consectetur dolore culpa deserunt quis officia do. Aliqua aliqua eu velit excepteur culpa ea dolore occaecat. Ea est cillum nulla ea cupidatat esse adipisicing ut. Cillum amet esse ex ex enim elit excepteur ullamco in qui aliqua Lorem veniam. Sunt consectetur aliquip commodo exercitation pariatur id tempor adipisicing aliqua. Fugiat cillum ea irure laborum consectetur ipsum. Excepteur laboris eu veniam ex ad consequat esse id ullamco ea cupidatat ad nisi. Aute nisi velit excepteur adipisicing dolore pariatur sunt. Dolore mollit cupidatat esse sint ipsum est nulla nisi eu cupidatat excepteur sint cillum. Culpa irure magna culpa eiusmod ut nisi ipsum veniam elit est occaecat veniam. Sunt adipisicing eu minim qui ea laboris ullamco ut. Aliquip occaecat in occaecat ut tempor dolore sunt consequat qui cillum reprehenderit. Voluptate commodo cupidatat ullamco sint aliquip veniam consequat do. Nisi velit excepteur aute laboris aliqua fugiat minim occaecat quis non incididunt.

**Keywords** — Precoding Techniques, High-Troughput Satellite Communications, MIMO, Outdated CSI

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[1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, “Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios,” *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

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# Symbols and Notations

## Abbreviations

| | |
|---|---|
| **acf** | autocorrelation function |
| **AR** | autoregressive |
| **AWGN** | additive white Gaussian noise |
| **BS** | base station |
| **BD** | block diagonalization |
| **BER** | bit error rate |
| **CS** | circularly symmetric |
| **CSI** | channel state information |
| **DFT** | discrete Fourier transform |
| **DPC** | dirty paper coding |
| **EVD** | eigenvalue decomposition |
| **FFT** | fast Fourier transform |
| **FIR** | finite impulse response |
| **GEO** | geostationary earth orbit |
| **GW** | gateway |
| **HTS** | high-throughput satellite |
| **IBR** | information bit rate |
| **i.i.d.** | independent and identically distributed |
| **KKT** | Karush-Kuhn-Tucker |
| **LEO** | low earth orbit |
| **LoS** | line-of-sight |
| **LSV** | left singular vector |

**MI** mutual information

**MIMO** multiple-input multiple-output

**MMSE** minimum mean squared error

**MSE** mean squared error

**MU** multi-user

**NLoS** non-line-of-sight

**pdf** probability density function

**PAM** pulse amplitude modulation

**PSD** power spectral density

**PSK** phase shift keying

**QAM** quadrature amplitude modulation

**QoS** Quality–of–Service

**RTT** round-trip time

**RX** receiver

**SINR** signal-to-interference-plus-noise ratio

**SISO** single-input single-output

**SNR** signal-to-noise ratio

**SR** sum rate

**SU** single-user

**SVD** singular value decomposition

**THP** Tomlinson–Harashima precoding

**TX** transmitter

**UT** user terminal

**WMSE** weighted mean squared error

**WMMSE** weighted minimum mean squared error

**WSR** weighted sum rate

**ZF** zero–forcing

## System Parameters

$K$ Number of UTs

| | |
|---|---|
| $N_T$ | Number of transmit antennas at the BS |
| $N_R$ | Number of receive antennas at each UT |
| $N_S$ | Number of data streams per UT |
| $P_t$ | Total transmit power |
| $N_0$ | Noise power on each receive antenna |

## Symbols

**SU–MIMO**

**MU–MIMO** TO DO

# Part I

# Background Material

# Part II

# Precoding for Satellite Communication Systems

# Chapter 5

# High-Throughput Satellite Communication Systems

The preceding chapters developed the theory of linear precoding for generic single-user (SU)- and MU-MIMO systems. Before turning to the specific problem addressed by this work, the present chapter situates these results within the context of satellite communications. We first motivate the use of precoding in modern high-throughput satellite (HTS) systems and describe the system architecture considered in the remainder of this work. We then discuss the channel features and modeling assumptions that distinguish the satellite setting from the terrestrial MU-MIMO scenario of Chapter 4. Finally, we review the main families of precoding techniques proposed in the literature and identify the specific category that is relevant for our work.

## 5.1 Motivation

Satellite communications have recently regained considerable attention as a key enabler of ubiquitous broadband connectivity. They provide cost-efficient coverage of large geographical areas, including regions where deploying a terrestrial infrastructure is impractical, and are envisioned as an integral component of future 5G and beyond networks [29, 30].

To meet the rapidly growing demand for data rates, modern satellite systems have evolved from single-beam architectures toward high-throughput satellite (HTS) systems, in which the coverage area is partitioned into many narrow spot beams. This multi-beam architecture enables spatial multiplexing by concurrently delivering independent data streams to geographically separated users. In conventional designs, inter-beam interference is managed through partial frequency reuse and polarization isolation, whereby adjacent beams are assigned non-overlapping frequency sub-bands or orthogonal polarizations. While effective at reducing interference, this approach significantly constrains spectral efficiency, since the full available bandwidth cannot be reused across all beams simultaneously. On the other hand, full frequency reuse maximizes the spectrum available per beam. The price to pay is a substantial increase in inter-beam interference: signals intended for one beam leak into adjacent beams through the sidelobes of the antenna radiation pattern, and the system transitions from a noise-limited regime to an interference-limited one.

Precoding has emerged as the natural solution: by exploiting channel state information (CSI) at the transmitter, interference can be suppressed while the available spectrum is reused across all beams [29].

## 5.2 Modeling Aspects

### System Architecture

An illustration of the system architecture of a multi-beam satellite system is shown in Figure 5.1. It consists of three main entities: the gateway (GW), the satellite or base station (BS), and the user terminals (UTs). The GW is connected to the core network and usually performs all baseband processing. The link between the GW and the satellite is referred to as the feeder link, while the link between the satellite and the UTs is referred to as the user link. The direction from the GW throught the BS to the UTs is called the forward link, while the opposite direction is called the return link. The forward link is by far the most rate-demanding direction, and centralizing the precoder design at the GW (or BS) aligns naturally with the MU-MIMO downlink framework developed in Chapter 4.

Throughout this work, we focus on the forward link of a multi-beam HTS system serving mobile UTs.

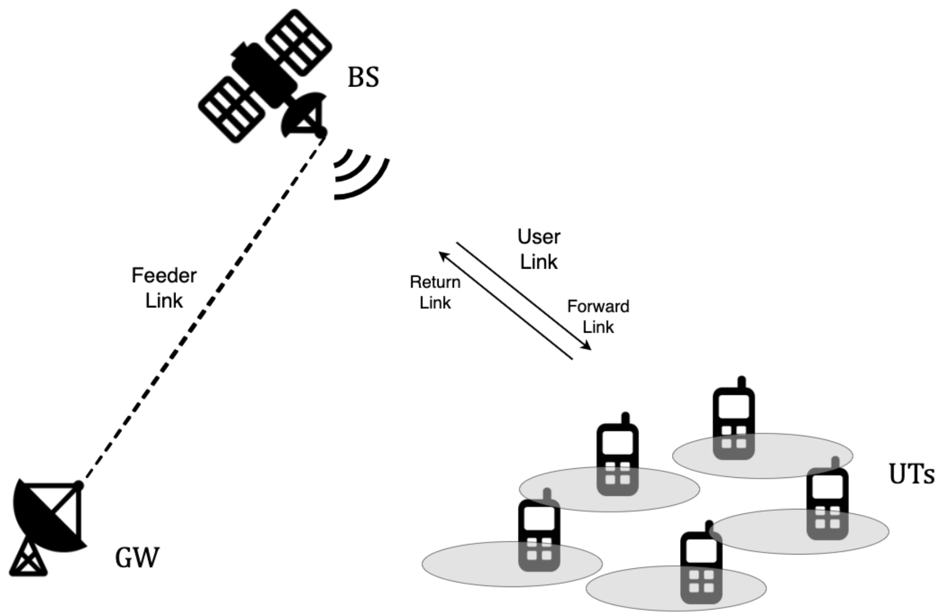



**Figure 5.1:** Schematic view of the multibeam HTS communication system architecture.

### Features of Satellite Communications

Compared to terrestrial MU-MIMO systems, satellite communications exhibit several distinctive features. The propagation conditions are dominated by a line-of-sight (LoS) component between the satellite and the UTs, accompanied by a weaker scattered contribution arising from the local environment of the UT [31]. The free-space path loss is several orders of magnitude larger than in terrestrial links, and the received power can be further reduced by atmospheric impairments such as rain attenuation. For mobile satellite services, however, communication often takes place in the L- or S-band, where propagation losses are smaller than in the Ka-band commonly used for fixed satellite services [29]. This comes at the expense of a more limited available bandwidth, which further motivates aggressive frequency reuse. For the present work, the two most relevant features are therefore the mobility of the UTs and the large propagation delay.

The mobility of the UTs and/or the satellite makes the channel time-varying through Doppler effects. For geostationary earth orbit (GEO) satellites, which appear essentially stationary relative to the Earth's surface, the satellite-induced Doppler shift is negligible, and the dominant source of channel variation is UT motion alone. For low earth orbit (LEO) satellites, the high orbital velocity produces a substantial satellite-induced Doppler shift. However, since the satellite's trajectory is known with high accuracy, this contribution is deterministic and can be pre-compensated (see e.g. [32]). The residual, more difficult component to track is the Doppler shift induced by the motion of the UT, since it depends on the instantaneous user velocity and propagation geometry. So, in both cases, UT mobility is the primary driver of channel time-variation, and it causes the channel to fluctuate at a rate governed by the UT speed and the carrier frequency.

The large geographical distances between the GW, the satellite, and the UTs introduce a propagation delay that is several orders of magnitude larger than in terrestrial systems. For a GEO satellite, which orbits at an altitude of approximately 35,786 km, the one-way propagation delay is on the order of 250 ms. Even for a LEO constellation, the round-trip time (RTT) remains in the tens of milliseconds. Since the precoder is computed based on CSI provided by the UTs, this delay implies that the channel used to design the precoder is inevitably outdated relative to the channel experienced when the precoded signal reaches the UTs. In a static scenario, this mismatch is of limited importance. In combination with a time-varying channel, however, it forms a major challenge for precoding design.

### Channel Model

No single probabilistic model captures the full range of propagation conditions encountered in mobile satellite communications, and a variety of models have been proposed in the literature for different scenarios.

In this work, we begin with a deliberately simplified model in which the channel of $\mathrm{UT}_k$ is represented as a frequency-flat Rician fading channel [31, 33]:

$$\mathbf{H}_k(t) = \sqrt{\frac{\kappa}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{LoS}}(t) + \sqrt{\frac{1}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{NLoS}}(t), \tag{5.1}$$

where $\kappa \geq 0$ is the Rician factor, $\mathbf{H}_k^{\mathrm{LoS}}(t)$ is the deterministic line-of-sight (LoS) component, and $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ is the time-varying stochastic non-line-of-sight (NLoS) component.

The Rician factor quantifies the power ratio between the LoS and the NLoS contribution. It typically takes values around 5 dB in urban environments and up to 15 dB or more in rural environments. In this work, it is set to 5 dB.

The LoS component is treated as time-invariant, and the small spatial variations across the antenna arrays are neglected. So, all entries of $\mathbf{H}_k^{\mathrm{LoS}}$ share a common, constant phase offset $\theta_k$ associated with UT $k$:

$$\mathbf{H}_k^{\mathrm{LoS}} = e^{j\,\theta_k}\,\mathbf{1}_{N_R \times N_T} \tag{5.2}$$

As a consequence, the LoS component does not contain any temporal or spatial variation.

The NLoS component $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ captures the scattered multipath contribution. Each entry is considered independent and identically distributed (i.i.d.), and modeled as a zero-mean unit-variance complex Gaussian random variable consistent with the Rayleigh fading model as described in Appendix C. The temporal correlation of the NLoS entries is governed by the Doppler spread induced by UT mobility. Under the classical isotropic scattering assumption, the power spectral density of each channel entry follows the Jakes model:

$$S_{\mathbf{H}_k^{\mathrm{NLoS}}}(f) = \frac{1}{\pi f_D \sqrt{1-(f/f_D)^2}}, \quad |f| \leq f_D, \tag{5.3}$$

where $f_D = v f_c / c$ is the maximum Doppler frequency, determined by the UT speed $v$, the carrier frequency $f_c$, and the speed of light $c$. Further details on the Doppler power spectrum are provided in Appendix C.3, and the generation of a time-correlated random process with a prescribed autocorrelation function or power spectral density is addressed in Appendix D.

The model as stated rests on two simplifying assumptions: spatial correlation between channel entries is neglected, and the LoS Doppler shift is assumed to be perfectly pre-compensated. While these assumptions yield a tractable and well-understood baseline, they do not always hold in practice. At the end of the following chapter, we therefore describe how this model can be extended to a more realistic setting that accounts for spatial correlation and a residual LoS Doppler component, and we briefly evaluate the impact of these features on the performance of the proposed precoding techniques.

## 5.3 Classification

A large number of precoding techniques have been proposed for satellite communications, differing in the assumptions they make about the system and in the design objectives they pursue. A comprehensive survey can be found in [34]. The objective of this section is not to reproduce this overview, but rather to position the present work within the broader landscape and to identify the techniques that are most closely related.

### A Taxonomy of Precoding Techniques

Besides the distinction between linear and nonlinear techniques, existing satellite precoding methods can be classified along several other non-exclusive axes. For each axis we briefly recall the main alternatives and indicate the choice made in this work.

A first distinction concerns the traffic model. In practical multibeam satellite systems, a single beam may simultaneously serve multiple UTs within the same frame, which leads naturally to a multi-group multicast formulation. To avoid additional complications introduced by multicast scheduling, we restrict the analysis to the unicast case, in which only a single UT is served per beam. A second axis concerns the optimization criterion. Depending on the system objective, the precoder may be designed to maximize the sum rate under a transmit power constraint, or to minimize the transmit power provided that Quality–of–Service (QoS) metric is met. Closely related to this is the choice of power constraint. In practical satellite payloads, each feed is typically driven by its own power amplifier, which makes per-feed power constraints the most natural formulation from an implementation perspective. However, owing to the additional complexity associated with per-feed power constraints, most existing precoding techniques are formulated under a total transmit-power constraint. In this dissertation, the emphasis is placed on sum-rate maximization subject to a total transmit power constraint, in line with the terrestrial MU-MIMO framework developed in the previous chapters. A further classification axis is the system architecture. In large multibeam systems, precoding may be distributed across multiple GWs, and part of the processing may even be transferred on board the satellite in order to relax feeder-link limitations. Here, the simplest architecture is considered: a single GW performs all precoding on the ground and serves the multibeam payload through a common feeder link.

Finally, and most importantly for the present work, precoding techniques differ in the type of CSI assumed at the transmitter. Three regimes are commonly distinguished: perfect instantaneous CSI, statistical CSI, and imperfect or outdated CSI. The vast majority of the satellite precoding literature assumes perfect instantaneous CSI at the GW. This assumption becomes unrealistic in the scenario considered here, where the combination of large propagation delay and a time-varying channel due to a UT-induced Doppler renders the CSI available at the GW systematically outdated. Precoding under outdated CSI is therefore the regime adopted in this work.

### Related Work and Positioning

To the best of our knowledge, only two works have addressed the design of linear precoding for mobile multi-beam satellite systems with outdated CSI under assumptions comparable to ours.

The first is the work of Vázquez et al. [35], which considers an L-band mobile interactive multi-beam system over a GEO satellite. The authors model the feedback loop as a single delay $\tau$ on the order of 700 milliseconds and adopt a simple closed-form minimum mean squared error (MMSE) precoder. Their key analytical observation is that, for a fixed precoder, the random phase fluctuations of the fading process affect the desired and the interfering terms symmetrically and therefore cancel in the resulting signal-to-interference-plus-noise ratio (SINR), only the amplitude fluctuations of the fading process remain visible. As a consequence, the ergodic sum rate is essentially unchanged by outdated CSI, and the dominant practical impairment becomes the mismatch between the SINR predicted by the outdated CSI and the SINR effectively experienced by the UTs, which is handled by fixed back-off margins in the link-adaptation stage.

The second is the work of Joroughi et al. [36], which considers a maritime multi-beam scenario at L-band. The channel is decomposed into a time-invariant beam-gain matrix and a time-varying diagonal attenuation matrix, and the authors propose two CSI feedback strategies. The first is a memoryless strategy that feeds the GW with the latest delayed snapshot of the attenuation.

The corresponding precoder is a conventional zero–forcing (ZF) or MMSE precoder computed from this estimate. The second is a memory-based strategy that averages the attenuation over several past snapshots, and it is paired with a novel two-stage precoder: a first stage suppresses inter-beam interference, while a second diagonal stage compensates for the residual time-varying attenuation.

While the works above tackle the same regime of outdated CSI in mobile multi-beam satellite systems, our approach differs in ...

# Chapter 6

# Precoding under Outdated CSI

In Chapter 4, linear precoding was developed under the implicit assumption that the CSI used to design the precoder coincides with the CSI experienced by the UTs at the moment of transmission. As discussed in Chapter 5, this assumption breaks down in the mobile satellite scenario considered in this work, owing to the combination of a large propagation delay and a UT-induced Doppler spread. This chapter is devoted to characterizing the impact of this mismatch on precoder performance and to developing a strategy that compensates for it.

Section 6.1 formulates the problem precisely and introduces the CSI feedback delay as the key operating parameter. From there, the chapter proceeds in three steps. First, Section 6.2 quantifies the impact of outdated CSI on the achievable rate of the linear precoders developed in Chapter 4. Section 6.3 then proposes a compensation method that exploits the temporal correlation of the channel in order to predict the instantaneous channel at the time of transmission. Finally, Section 6.4 re-evaluates both the impact of outdated CSI and the proposed compensation strategy under a more realistic channel model.

To end this work, the results are applied to a set of representative example scenarios, after which the main findings are summarized in a final conclusion.

## 6.1 Problem Formulation

We consider the forward link of the multi-beam HTS system as described in Chapter 5. The GW serves $K$ UTs simultaneously through the satellite, equipped with $N_T$ feeds, by applying a linear precoder to the data symbols. Each UT is equipped with $N_R$ receive antennas. The channel of UT $k$ at time $t$ is denoted by $\mathbf{H}_k(t) \in \mathbb{C}^{N_R \times N_T}$ and follows the Rician model from (5.1). The received signal at UT $k$ reads

$$\mathbf{y}_k(t) = \mathbf{H}_k(t)\, \mathbf{F}(t)\, \mathbf{a}(t) + \mathbf{n}_k(t), \tag{6.1}$$

where $\mathbf{a}(t)$ collects the data symbols of each stream intended for UT $k$ into a column vector, and $\mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_{N_R})$ is additive noise. The precoder satisfies the total transmit-power constraint $\mathrm{tr}\big(\mathbf{F}(t)\mathbf{F}^{\mathsf{H}}(t)\big) \leq P_T$ at any moment in time.

In practice, the channel must first be estimated at the UTs and fed back to the GW, before it can be used to design the precoder. Denoting the aggregate elapsed time by the CSI feedback delay $\tau_{\text{CSI}}$, the precoder applied at time $t$ is necessarily based on the outdated channel state $\mathbf{H}(t-\tau_{\text{CSI}})$. How severely this mismatch degrades performance depends on how much the channel has evolved during $\tau_{\text{CSI}}$.

The temporal variability of the channel is characterised through the normalised autocorrelation function of its NLoS component, denoted by $R_h(\tau)$. It is shown in Appendix C.3 that, under Jakes' model, the autocorrelation between two channel realizations separated by a lag $\tau$ equals the zeroth-order Bessel function of the first kind, evaluated in $2\pi f_D \tau$. The coherence time of the NLoS component is defined as the lag at which the autocorrelation first falls to one half of its peak value:

$$T_c \triangleq \max\{\tau > 0 \, : \, R_h(\tau) \geq \frac{1}{2}\}. \tag{6.2}$$

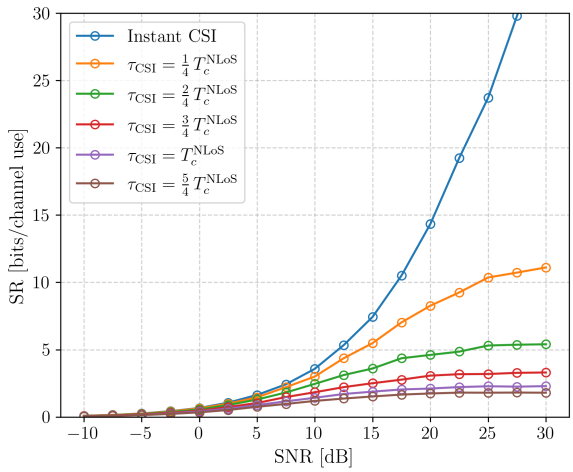
Rather than studying the impact of $\tau_{\text{CSI}}$ in absolute terms, the CSI feedback delay is expressed as a fraction of the coherence time throughout the analysis. Two operating points sharing the same $\tau_{\text{CSI}}/T_c$ exhibit, by construction, the same correlation between the outdated and the instantaneous channel, irrespective of the absolute values of $\tau_{\text{CSI}}$ and $T_c$. Reporting results as a function of $\tau_{\text{CSI}}/T_c$ therefore yields conclusions that apply to any combination of carrier frequency, UT velocity and orbital altitude consistent with the underlying Rician model.

## 6.2 Impact of Outdated CSI

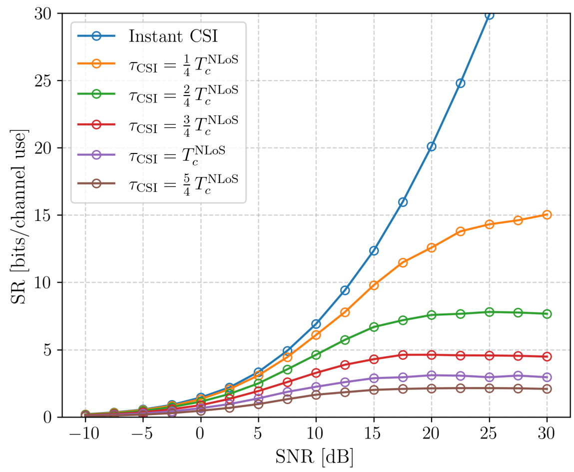
In order to quantify the impact of outdated CSI on the system performance, the achievable rate obtained by linear precoders developed in Chapter 4 is simulated as a function of the signal-to-noise ratio (SNR) for a system in which a BS with 8 transmit antennas serves 4 UTs with 2 receive antennas each. Each precoder is designed based on the outdated channel state $\mathbf{H}(t-\tau_{\text{CSI}})$ and applied to the instantaneous channel $\mathbf{H}(t)$. The resulting sum rate is averaged over more than 100 channel realizations, and over multiple coherence periods of the fading process for each realization. Figures 6.1 and 6.2 report the results for a CSI feedback delay $\tau_{\text{CSI}}$ that varies from $0\,T_c$ (instantaneous CSI, used as a reference) up to $5/4\,T_c$, slightly beyond the coherence time.

All three precoders are very sensitive to outdated CSI. Even a delay of only a quarter of the coherence time causes a substantial loss on the achievable rate at moderate and high SNR values. The loss with respect to instantaneous CSI grows monotonically with $\tau_{\text{CSI}}/T_c$, confirming the intuition that the more outdated the CSI, the worse the performance. Once the CSI feedback delay surpasses the coherence time of the NLoS component, the system becomes essentially unusable. This aligns with the definition of $T_c$. Beyond the point $\tau_{\text{CSI}} \geq T_c$, the channel available at the GW and the channel actually experienced by the UTs are only weakly correlated such that the precoder is in effect designed for a channel that bears little resemblance to the true one. As a consequence, the precoder fails to exploit the spatial dimension of the channel and the sum-rate collapses.

A closer look at Figure 6.1 shows that both ZF and block diagonalization (BD) precoding saturate at high SNR once the CSI is even slightly outdated. Both ZF and BD precoders completely eliminate the interference between the users, no matter the ratio of the transmit power to the noise power. With outdated CSI, the precoder nulls interference along the wrong directions, leaving a residual inter-user interference that is proportional to the transmit power. As the SNR increases, the system transitions from a noise-limited regime to an interference-limited one and the desired signal and the residual interference grow at the same rate. Therefore, the SINR saturates and the achievable rate converges to a finite ceiling, that is determined solely by the magnitude of the channel evolution over the feedback delay.




**(a)** ZF Precoding




**(b)** BD Precoding

**Figure 6.1:** The achievable sum-rate of a $8 \times \{2, 2, 2, 2\}$ MU-MIMO system as a function of the SNR, for ZF and BD precoding. The different curves correspond to different values of the CSI feedback delay $\tau_{\text{CSI}}$, expressed as a fraction of the coherence time $T_c$.

The weighted minimum mean squared error (WMMSE) precoder exhibits a qualitatively different behaviour. The achievable rate reaches a maximum and decreases slightly beyond it when the precoder design relies on outdated CSI, as is clearly visible in Figure 6.2. Unlike ZF and BD, WMMSE precoding adapts its solution to the operating SNR. At low SNR it trades interference suppression against desired-signal power, while at high SNR the regularization vanishes and it converges to the interference-free BD solution. The algorithm then increasingly concentrates power in directions that it expects to be beneficial, but which, due to the channel mismatch, actively generate inter-user interference in the actual channel. The ZF-type precoders do not exhibit this behaviour because they steer in the wrong direction just as aggressive for any SNR value.

A final observation concerns the reference curve obtained with instantaneous CSI. Even in this idealised case, the achievable rate lies noticeably below the values obtained in Chapter 4 under a purely Rayleigh fading channel. This accounts for all three precoding techniques, and is a direct consequence of the LoS contribution in a Rician channel. Indeed, the LoS component is rank-one, such that the largest fraction of the channel energy is concentrated along a single spatial direction. A purely Rayleigh fading channel, by contrast, provides rich spatial diversity and a well-conditioned channel matrix, granting the precoder significantly more spatial multiplexing

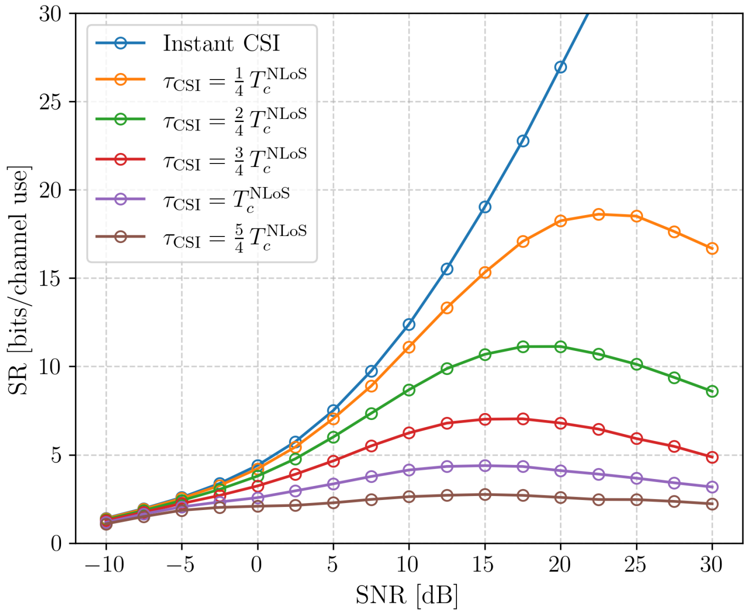



**Figure 6.2:** Similar figure as Fig. 6.1, but for WMMSE precoding.

gain. Therefore, the LoS component imposes an inherent limit on the spatial degrees of freedom available, regardless of the quality of the CSI.

Taken together, these results show that even a small feedback delay poses a threat to the achievable rate at moderate and high SNR, and that no choice of precoder can restore the linear growth observed with instantaneous CSI by itself. The next section therefore turns to the design of a compensation strategy that targets the source of the mismatch, which is the temporal mismatch between $\mathbf{H}(t - \tau_{\text{CSI}})$ and $\mathbf{H}(t)$, rather than its consequences.

## 6.3 Compensation for Outdated CSI

intro to the section

### 6.3.1 Idea

- explain the idea: use an AR model to predict the CSI for the time instant of transmission, and send the predicted version to the GW. The GW acts as if nothing happened, all extra compensation processing happens at the UTs. say what an AR model does, but do not go into detail, refer to Appendix E for more info (derivation Yule-Walker equations, ...)

- explain the assumptions the UTs have access to the statisctic of the channel further, we will assume that the rate of pilot messages is not constraint but free to choose.

### 6.3.2 AR parameter optimization

- explain the free parameters of the AR model the pilot message rate. this corresponds to the frequency of the taps of the AR model. the context window length. this determines the size of the memory at the UT, and how much in the past we look te predict the future channel impulse response.

- shortly go through the optimization process. instead of a full grid search over the window length, the order of the ar-model and the delay of the CSI feedback (  propagation delay), we first choose a fixed number of taps, and then fix the number of tabs to that number. Then, we choose the best window length.

we start with a window length of 1.5 Tc. This is a reasonable choice. We do not expect a lot of useful information to be lost with this length. plot the achievable rate with compensation as a function of the number of tabs (with a fixed window length of 1.5). the number of tabs the model uses is directly correlated to the pilot message rate. (number of tabs equals pilot message rate times the context window length) hypothesis: the higher the pilot range, the better the prediction quality (more information accessible) the hypothesis is correct. Figure 6.3 shows that using to few taps can drastically decrease the quality of the prediction and therefore the achievable sum-rate of the system when using the compensation technique compared to more taps. The use of the 6 pilot messages per coherence times seems ideal. There is even a slight decrease in performance when using more taps.

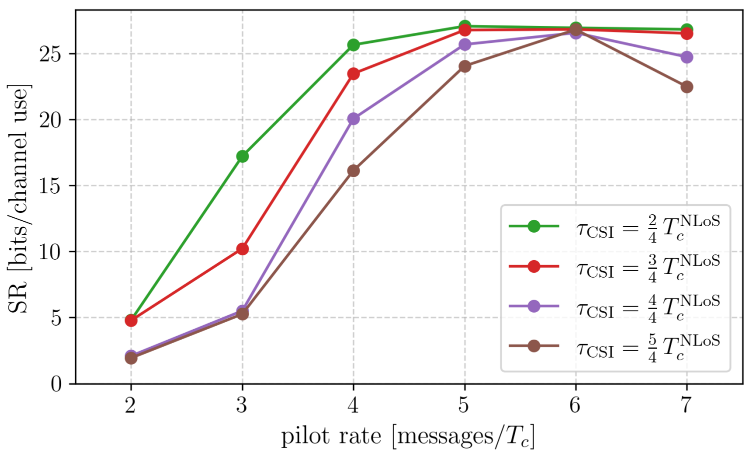



**Figure 6.3:** The impact of the pilot message rate on the achievable sum-rate of a $8 \times \{2, 2, 2, 2\}$ system using WMMSE precoding at an operating point with an SNR of 20 **dB**. The different curves represent different CSI feedback delays. The context window length is fixed to $1.5\,T_c$.

next, we optimize the context window length, which corresponds to the memory size of the autoregressive (AR)-model. hypothesis: the longer the context window, the better the prediction and thus the system performance. A longer window will not harm the performance, since it only adds extra (useless) information. the hypothesis gets acknowledged by figure 6.4. The figure shows that a too short context window length can drastically decrease the achievable sum-rate. Once the context window exceeds 1 Tc, no additional increase in achievable sum-rate is observed for the plotted values of CSI feedback delay. We choose the context window length equal to 1 Tc

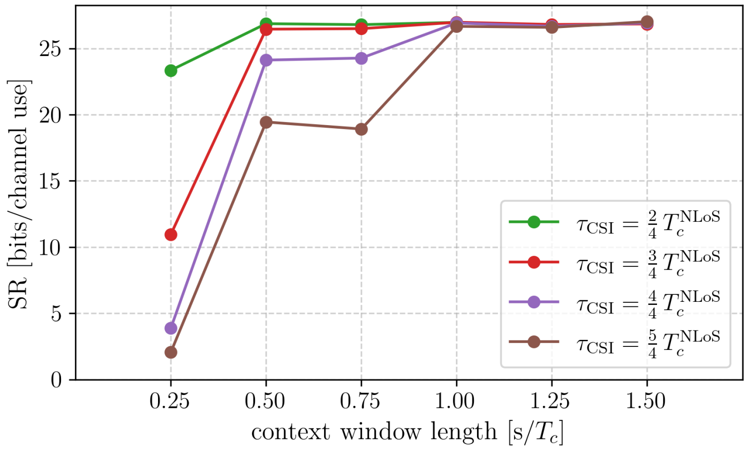
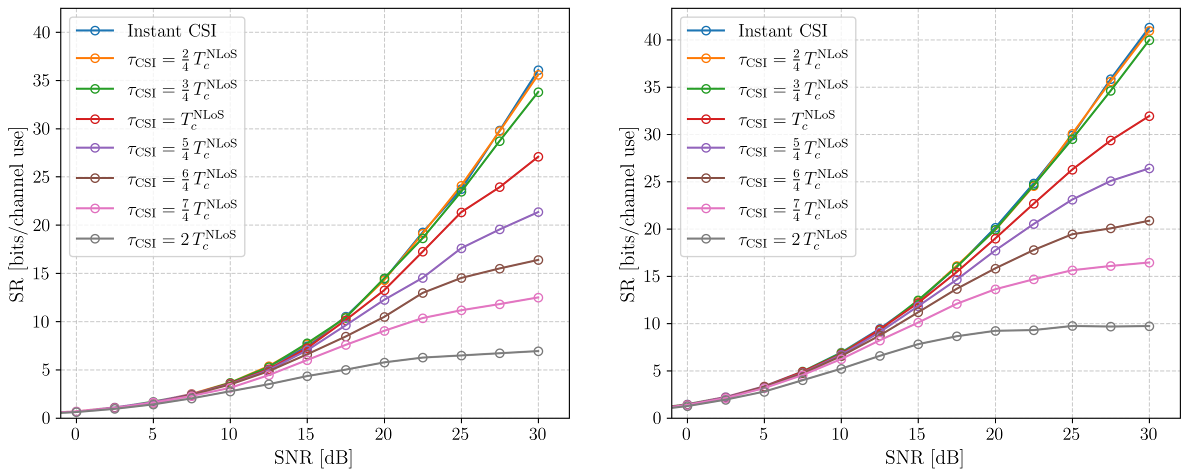



**Figure 6.4:** The impact of context window length of the AR-model on the achievable sum-rate of a $8 \times \{2, 2, 2, 2\}$ system using WMMSE precoding sing WMMSE precoding at an operating point with an SNR of 20 **dB**. The different curves represent different CSI feedback delays. The number of taps id fixed to 6.




**(a)** ZF Precoding

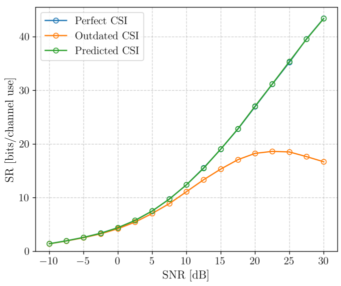
**(b)** BD Precoding

**Figure 6.5:** TO DO

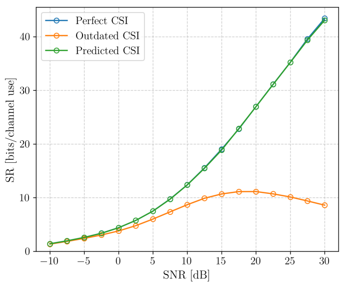
### 6.3.3 Simulation Results

## 6.4 Evaluation on a More Realistic Channel

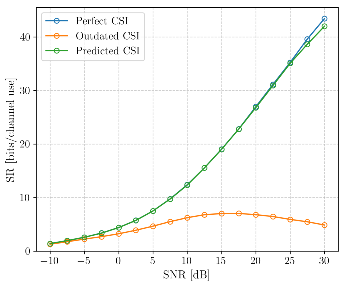
## 6.5 Conclusion



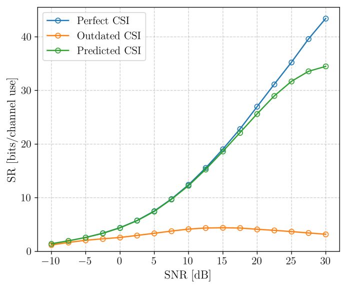

(a) $\tau_{\text{CSI}} = \frac{1}{4}T_c$



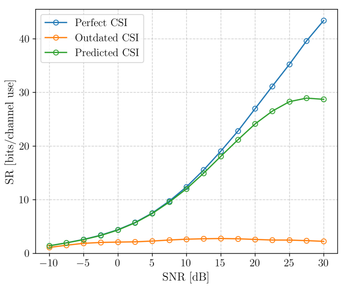

(b) $\tau_{\text{CSI}} = \frac{2}{4}T_c$



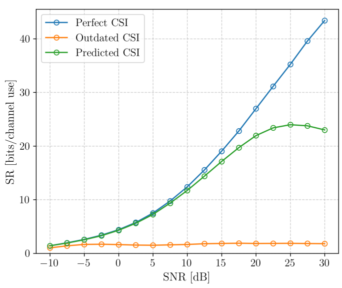

(c) $\tau_{\text{CSI}} = \frac{3}{4}T_c$



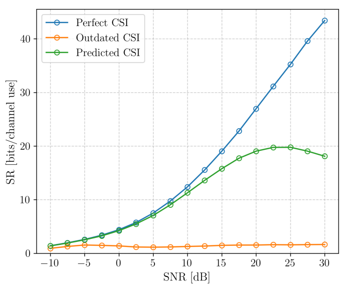

(d) $\tau_{\text{CSI}} = \frac{4}{4}T_c$



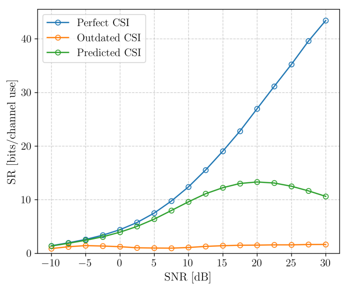

(e) $\tau_{\text{CSI}} = \frac{5}{4}T_c$




(f) $\tau_{\text{CSI}} = \frac{6}{4}T_c$




(g) $\tau_{\text{CSI}} = \frac{7}{4}T_c$




(h) $\tau_{\text{CSI}} = \frac{8}{4}T_c$

**Figure 6.6:** TODO

# Part III

# Appendices

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